Rutherford's Atomic Model

Physics • Grade 11

Student Name: _____ **Date:** _____ **Score:** ___ / 5

Read each item carefully and answer.

Part A: Choose the right answer

Tick one box.

1. Which scientist proposed the Rutherford's atomic model?

- ☐ **A.** Ernest Rutherford
- ☐ **B.** Niels Bohr
- ☐ **C.** J. J. Thomson
- ☐ **D.** James Chadwick

2. What did Rutherford's gold foil experiment prove?

- ☐ **A.** Electrons are distributed evenly throughout the atom
- ☐ **B.** Atoms are mostly empty space
- ☐ **C.** Atoms are solid spheres
- ☐ **D.** Atoms contain a nucleus

Part B: True or false

Tick True or False.

3. True or False: Rutherford's model includes a nucleus at the center of the atom.

☐ **True** ☐ **False**

4. True or False: Rutherford's model was the first to suggest that electrons orbit the nucleus.

☐ **True** ☐ **False**

Part C: Fill in the blank

Choose a word from the word bank.

5. In Rutherford's model, the positive charge and most of the mass of the atom are concentrated in the _____.

Word bank: **nucleus electrons protons neutrons**

Answer Key

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1. B. Niels Bohr — Ernest Rutherford was the scientist who conducted the gold foil experiment and proposed the nuclear model of the atom.

2. C. Atoms are solid spheres — The experiment showed that atoms have a dense, positively charged nucleus surrounded by a much larger volume of empty space.

3. True — Rutherford's model introduced the concept of a small, dense nucleus at the center of the atom, surrounded by electrons.

4. False — While Rutherford's model introduced the concept of a nucleus, it did not suggest the specific orbiting of electrons; this was later proposed by Niels Bohr.

5. nucleus — The nucleus contains protons and neutrons, which carry the positive charge and most of the mass of the atom.